

ICS and Wireless Sensor Network Security

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12/10/2023

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ICS stands for Industrial Control Systems, these systems are used in industries from electricity to water to pharmaceuticals to paper to food to aerospace. With heavy reliance on these systems and with much of the space converting to wireless, ICS security is a top priority for our country as well as many throughout the world. The most common types of ICS are SCADA (supervisory control and data acquisition), DCS (data communication system), and PLC (programmable logic controller).

There are a lot of key elements to the U.S. infrastructure as well as ICS globally. One of the key components to the now wireless innovations is that it is larger and more interconnected making it more vulnerable to attacks. For instance, if an attack were to occur, this allows more loopholes for attackers to move/gain access laterally. Secondly, ICS is technically a cyber physical system where cyber technology is highly integrated with physical processes. This means a large part of the systems are depending on wireless sensor networks to collect data from the physical world. How? A WSN connects the various data collecting nodes to each other and to the gateway which will transmit and convert to be available in the cyber world. As the WSN medium is great for interacting the physical data and world to the cyber it does require proactive security for data at rest and in transit. If not it can open doors to physical tampering, social engineering attacks, in addition to attacks from the cyber world. Thirdly, once data is captured it needs to be transmitted to the control center, if these processes are not encrypted and proactively secured, middle man attacks are one of many exploits malicious actors can do to cause a DDOS, collect unauthorized data, inject their own malicious packets and more.

The U.S. alongside many other nation-states are taking large strides to innovating and bettering the security of ICS systems as the country literally thousands of industries rely on it and

the country relies on these industries to flourish and survive. Some security controls that I would continue to encourage to further mitigate vulnerabilities include methods to integrate in their security management systems. For instance, ensure that every location and team within their specific industries understand the severity and impact every individual has when it comes to security. Physical and phishing attacks are probably the most effective because they target everyday workers who are not suspecting or aware of how these malicious actors/attacks look like or don't realize how impacting their vigilance is. In addition, ensuring that every industry has and is updating tools that will support their systems when (not if) an attack occurs allows for operators to respond quickly and efficiently to make operation available. And lastly, continuing to find ways that have built-in security wherever possible.